CHAPTER 88 Kidney transplantation and the principles of transplantation

Learning objectives

To recognise and understand:

- The immunological basis of allograft rejection
- The principles of immunosuppressive therapy
- The side effects of immunosuppressive therapy
- The major issues concerning organ donation
- The main indications for organ transplantation
- The surgical principles of organ transplantation
- The expected outcomes after transplantation
- The potential future developments in transplantation

INTRODUCTION

Successful solid organ transplantation represents one of the great medical advances of the twentieth century. The field continues to be an exciting and fast-moving one. Unfortunately, there continues to be a shortage of suitable donor organs for transplantation. In the UK there are approximately 4000 patients waiting for a kidney but only 3750 transplants are performed annually. This has led to a median waiting time for transplantation of around 3 years. Similar shortages exist for heart, lung, liver and pancreas transplantation across the world. Most transplant organs are from deceased donors, of which there are two types: donation after brainstem death (DBD) and donation after circulatory death (DCD). Living donation is limited to kidney, liver and lung transplantation.

ORGAN DONATION

Donation after brainstem death

Brainstem death occurs after severe brain injury as a result of either trauma or a cerebrovascular accident. Potential DBD donors are in an apnoeic coma that requires mechanical ventilation on the intensive care unit (ICU). There must be a known cause of irreversible brain damage demonstrated by a computerised tomography (CT) scan. Brainstem death is defined as the permanent loss of the capacity for consciousness and spontaneous breathing. These two essential functions are controlled by the brainstem. The reticular activating formation controls consciousness and is diffuse throughout the brainstem; the respiratory centre is in the medulla oblongata.

In this situation the circulation can be maintained for a period of time after death and DBD donors were formerly called heart-beating donors. In most countries it is accepted that brainstem death equates in medical, legal and religious terms to the death of the patient.

Before brainstem death testing can be considered, all reversible causes of coma must be excluded. These include: hypothermia, muscle relaxants, drugs with central nervous system depressive effects, alcohol intoxication, hypothyroidism, uraemic encephalopathy, hepatic encephalopathy, hypoglycaemia and hyponatraemia. When these preconditions are met then formal brainstem death testing can be undertaken (*Table 88.1*).

The UK guidelines state that the tests should be performed twice by two clinicians who are independent of the transplant team. One of them should be a consultant and the other must have been registered with the General Medical Council for at least 5 years.

TABLE 88.1 Clinical testing for brainstem death.

Cranial nerve reflexes	**1.** Pupillary reflex (cranial nerves II and III) **2.** Corneal reflex (cranial nerves V_1 and VII) **3.** Oculocephalic (doll's eyes movements – cranial nerves III, IV and VI) **4.** Vestibulo-ocular (cranial nerves VIII, III and VI) **5.** Cough/gag reflex (cranial nerves IX and X) **6.** It is not possible to test cranial nerves I and XII in unconscious individuals
Motor response	Absence of cranial nerve motor response to supraorbital pressure (cranial nerves V and VII) and absence of motor response in the cranial nerve distribution to adequate stimulation of any somatic area (commonly nail-bed pressure)
Apnoea test	After preoxygenation with 100% oxygen the patient is disconnected from the ventilator for >5 minutes in order to achieve a $PaCO_2$ ≥6.0 kPa (and pH<7.4). If there are no respiratory movements in response to the progressive hypercarbia then the test confirms that the brainstem respiratory centre has been destroyed. To prevent hypoxia during the apnoeic period, O_2 (6 L/min) is delivered via an endotracheal catheter

$PaCO_2$, partial pressure of carbon dioxide.

In DBD donors the organs continue to be perfused with oxygenated blood during the procurement surgery. The thoracic and abdominal organs are accessed through a median sternotomy and a midline laparotomy incision and their inflow vessels are cannulated. The organs to be recovered are usually dissected and mobilised with the heart beating. Ice-cold cardioplegia solution is then perfused into the coronary arteries via a cannula in the ascending aorta to stop the heart. At the same time the abdominal organs are perfused with ice-cold preservation solution via cannulae in the abdominal aorta and the portal vein. Each organ is removed and stored in fresh preservation solution and then packed in crushed ice for transport.

Donation after circulatory death

Donation after circulatory death describes the recovery of organs for transplantation after death confirmed by circulatory criteria. These donors were formerly called asystolic or non-heart-beating donors. There have been very significant increases in DCD programmes in many countries over the last decade. The modified Maastricht classification is widely used to categorise DCD (*Table 88.2*). Organ donation after unexpected and irreversible cardiac arrest is referred to as uncontrolled DCD. Donation after death resulting from the planned withdrawal of life-sustaining cardiorespiratory support is called controlled DCD.

In DCD donors cardiorespiratory arrest occurs prior to starting organ retrieval. The organs are therefore warm but not being perfused with oxygenated blood for a period of time before they are flushed with cold preservation solution. This warm ischaemia period should be limited as much as possible.

Controlled DCD donors are ICU-based and have suffered massive and irreversible cerebral damage but have an intact brainstem so that they are self-ventilating. In a situation where further attempts at treatment would be futile, the withdrawal of supportive treatment inevitably leads to cardiorespiratory arrest, and this usually occurs within a short time. In the UK, after cardiac arrest there is a mandatory 'no-touch' period of 5 minutes. This is deemed to be the time beyond which there is irreversible loss of cardiac and cerebral function. The donor is transferred from ICU to the operating department and a rapid median sternotomy and midline laparotomy are performed. The ascending aorta and abdominal aorta are cannulated and the organs are perfused with ice-cold preservation fluid *in situ* without any initial dissection. The warm ischaemic period is usually less than 10 minutes. Organ procurement is then carried out in standard fashion.

Uncontrolled DCD donors have usually suffered an unexpected and irrecoverable cardiac event either outside or inside hospital. After a period of attempted, but failed, cardiopulmonary resuscitation (CPR) and observation of the 5-minute rule, CPR with administration of high-concentration oxygen is recommenced, often using a mechanical resuscitation device. *In situ* renal cooling is performed by placing a double-balloon, triple-lumen perfusion catheter into the aorta via a femoral artery cut-down (*Figure 88.1*). The donor can then be transferred to the operating theatre and the kidneys are removed using standard techniques. Uncontrolled DCD donation is usually reserved for the kidneys only as they are able to recover from warm ischaemic periods of up to 45 minutes.

TABLE 88.2 Modified Maastricht classification of donation after circulatory death (DCD).

Category	Description	Type of DCD	Location
I	Dead on arrival at hospital	Uncontrolled	ED
II	Unsuccessful resuscitation after cardiac arrest	Uncontrolled	ED
III	Anticipated cardiac arrest after withdrawal of support	Controlled	ICU
IV	Cardiac arrest in brain-dead donor	Controlled	ICU
V	Unexpected cardiac arrest in ICU	Uncontrolled	ICU

ED, emergency department; ICU, intensive care unit.

Evaluation of the deceased donor

The absolute contraindications to organ donation include active systemic sepsis and transmissible infection. Malignancy within the last 5 years is also an absolute contraindication with the exception of tumours that do not metastasise (primary brain tumours, non-melanotic skin cancer and *in situ* carcinoma of the cervix). However, there is now evidence that organs from high-risk donors can be transplanted safely and

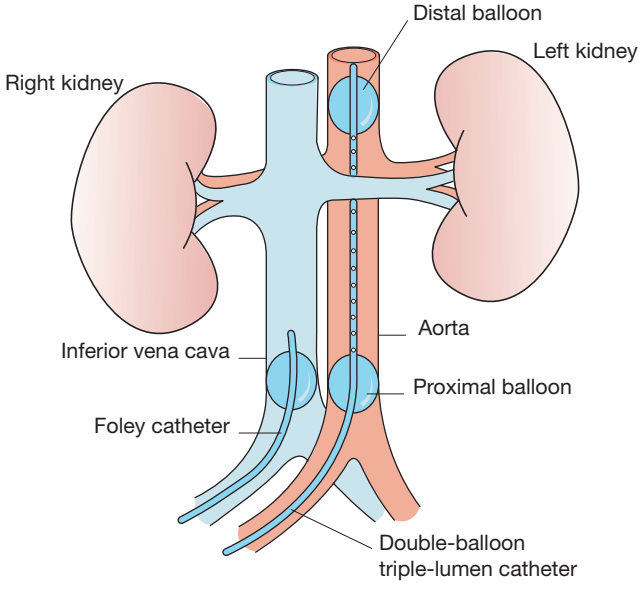

Figure 88.1 *In situ* perfusion of kidneys in a non-heart-beating donor (donation after circulatory death [DCD]). A double-balloon aortic catheter is introduced through a groin incision and 10–15 litres of chilled preservation solution is administered. The perfusate is vented through a Foley catheter introduced into the femoral vein.

Frederic Eugene Basil Foley, 1891–1966, urologist, Ancker Hospital, St Paul, MN, USA.

yield transplant outcomes that are equivalent to those from conventional donors. Thus, as a further response to the organ donor shortage, organs are now being transplanted from donors with meningitis/encephalitis, human immunodeficiency virus (HIV), hepatitis B and C and high-risk behaviour with the potential for blood-borne infection.

Living donation

Living kidney donation is possible because most individuals have two healthy kidneys and it is possible to live a normal life with a single kidney. Parts of non-paired organs can also be removed from live donors; these include liver and lung lobes, the tail of the pancreas and segments of small intestine.

The majority of liver transplants performed in India are from live donors and the number of programmes has expanded rapidly in recent years. The liver has the capacity to regenerate following resection of a segment or lobe. The growth of new liver tissue happens quickly and the liver returns to its pre-resectional mass. Liver resection is difficult because of the complex segmental anatomy (*Figure 88.2*). In children, the left lateral segment (segments II and III) of an adult provides enough liver function and this is a relatively straightforward procedure. In adult-to-adult live donor liver transplantation the whole right lobe of the liver is removed from the donor (segments V–VIII). This is a more complex operation and the risk of donor mortality is 0.5–1%.

Transplants from live donors have a number of advantages over deceased donor organs. Live donor organs are from healthy individuals without the comorbidities commonly seen in deceased donors. In the agonal period before death deceased donors exhibit marked changes in physiology related to a catecholamine storm and this can cause organ dysfunction. Clearly, live donor organs are not subjected to this insult. Potential live donors undergo a rigorous assessment process that includes imaging of the relevant vasculature by CT or magnetic resonance angiography (MRA) and tests of organ functional capacity. The operations are planned elective procedures undertaken in daylight hours and often adjacent theatres so that the cold ischaemia time is very short. It is usual for live donor organs to function immediately after transplantation and this is essential for liver and lung transplants. As a result of these advantages the long-term outcomes of live donor transplantation are superior to deceased donor outcomes.

Set against these advantages for the recipient is the fact that the live donor is subjected to a major operation that they do not need. All live donor operations have uncommon but potentially life-threatening morbidity rates and there is a small risk of death. The ethical issues raised by live donation are understandably complex.

ORGAN PRESERVATION

Transplant organs need to be stored and preserved in the period between procurement from the donor and transplantation into the recipient. Static cold storage is the traditional method of organ preservation, but more recently there have been developments in preservation by machine perfusion.

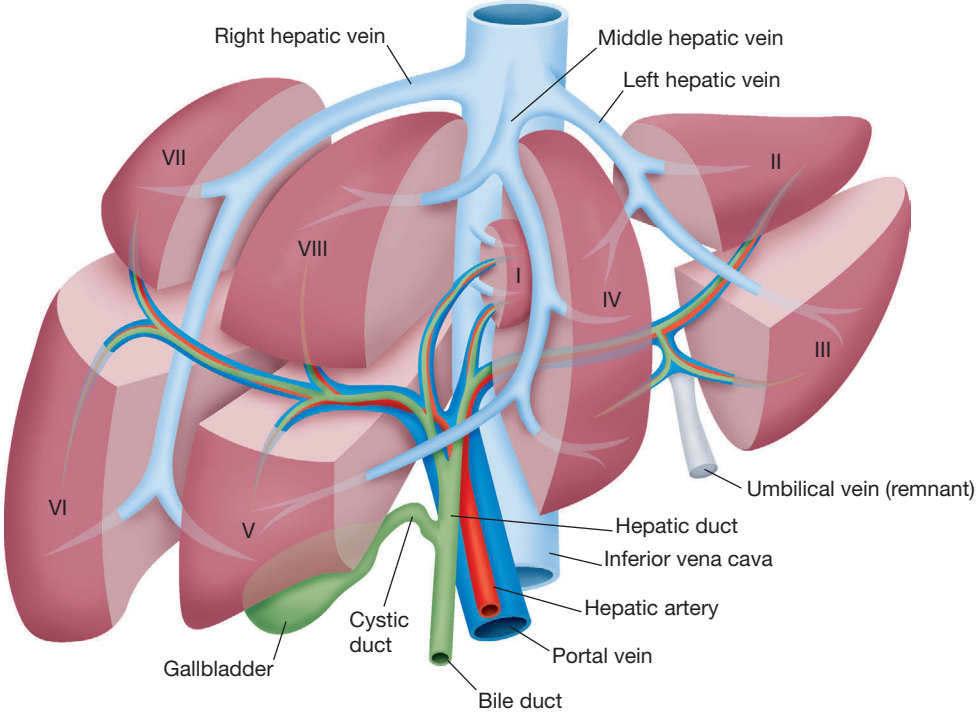

Figure 88.2 Segmental anatomy of the liver (Couinaud segments).

Claude Couinaud, 1922–2008, French surgeon and anatomist, described the segmental anatomy of the liver in his seminal book *Le Foie: Études anatomiques et chirurgicales*.

Pathophysiology of ischaemia

In the absence of oxygen, organs switch from aerobic to anaerobic metabolism. The mitochondrial electron transport chain cannot function without oxygen as the final electron acceptor. Oxidative phosphorylation ceases and, in consequence, so does the tricarboxylic acid (Krebs) cycle and the link reaction (conversion of pyruvate to acetyl coenzyme A). This leaves glycolysis as the only source of adenosine triphosphate (ATP). Anaerobic metabolism of one glucose molecule produces only two ATP molecules compared with the 36 ATP molecules that are produced during aerobic metabolism. Consumption of ATP rapidly exceeds production, leading to depletion of cellular ATP. Na^+/K^+ ATPase pumps are disabled and this leads to an influx of Na^+ ions into the cell down their concentration gradient. Na^+ ions are followed by H_2O by osmosis and this causes swelling and disruption of membrane-bound intracellular organelles and lysis of the cell membrane. Anaerobic glycolysis generates lactate and promotes intracellular acidosis. At low pH phospholipase and protease enzymes are activated and cause lysosomal damage and eventually cell death. ATP-dependent active transport of calcium out of cells is impeded, leading to the intracellular accumulation of Ca^{2+} ions and activation of the calcium-dependent processes. This leads to breakdown of the cytoskeleton and loss of cell structure.

Static cold storage

Hypothermia suppresses metabolism to maintain organ viability. The first requirement is to flush the donor organs with an appropriate preservation solution at a temperature of approximately 4°C (***Figure 88.3***). This is normally done *in situ* via cannulae placed in the donor aorta and portal vein. This process has three effects: (i) it flushes blood out of the microcirculation to prevent thrombosis; (ii) it cools the organs to a temperature of <5°C and so reduces tissue oxygen requirements; (iii) it replaces the normal extracellular fluid with the preservation fluid.

Figure 88.3 After removal from the donor, the kidney is flushed with chilled organ preservation solution and, if necessary, stored briefly on ice until transplanted into the recipient.

Preservation solutions have three main components: electrolytes, an impermeant and a buffer. In general, the electrolyte composition mimics the high K^+ ion concentration and low Na^+ ion concentration of intracellular fluid. This eliminates ionic fluxes, and therefore the movement of H_2O across cell membranes. Impermeants such as mannitol, lactobionate and raffinose are large osmotically active molecules. They cannot pass through the cell membrane and remain in the extracellular space. Here, they prevent cellular swelling by counteracting the osmotic force from intracellular proteins. The buffer component of preservation fluids is based on either HCO_3^- or PO_4^{3-} and acts to maintain a stable physiological pH in the extracellular space. A number of different preservation solutions are available, but the University of Wisconsin (UW) solution is widely regarded as the current gold standard.

Hypothermic machine perfusion

The transplant organ is placed in a sterile chamber and cold preservation fluid is continually recirculated through the vasculature at low temperature and pressure (30 mmHg) (***Figure 88.4***). The perfusion fluid is based on UW solution and some systems include oxygenation. This technique flushes the microcirculation more effectively. There are trial and meta-analysis data showing that hypothermic machine perfusion reduces the rate of delayed graft function (DGF) in renal transplantation, but the effect size is small.

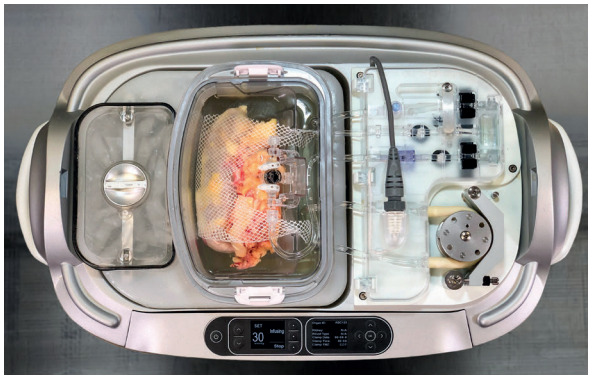

Figure 88.4 The LifePort® hypothermic perfusion machine (LifePort® Kidney Transporter courtesy of Organ Recovery Systems).

Normothermic machine perfusion

Normothermic machine perfusion (NMP) utilises the principles of cardiopulmonary bypass technology and has been used for heart, lung, liver and kidney preservation. Warmed and oxygenated red blood cell-based perfusate is circulated through the donor organ (***Figure 88.5***). This provides a more physiological environment that restores graft function *ex vivo* and replenishes depleted ATP levels. NMP restores organ function *ex vivo* and this allows for assessment of allograft quality and viability. This has led to increased utility of marginal organs that would previously have been discarded on the basis of adverse clinical parameters. NMP may also improve early

Sir Hans Adolf Krebs, 1900–1981, Professor of Biochemistry, University of Oxford, Oxford, UK.

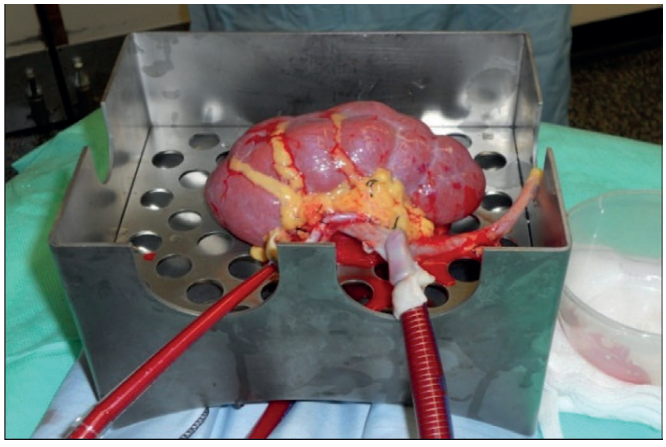

Figure 88.5 Donor kidney undergoing normothermic machine perfusion.

allograft function and is being developed as a platform for the delivery of pretransplant therapies.

Normothermic regional perfusion

This technique has been developed for donation after circulatory arrest. A cardiopulmonary bypass system is used to recirculate the donor's own blood through the thoracic and abdominal organs for 2 hours. The underlying principle is to avoid early cold ischaemia injury. The early experience of normothermic regional perfusion in DCD liver preservation and transplantation shows reduced rates of early allograft dysfunction.

IMMUNOLOGY OF TRANSPLANT REJECTION

ABO blood groups

ABO blood group antigens are glycoproteins with different carbohydrate components. ABO antigens are expressed not only on the surface of red blood cells but also on endothelial cells. In all organ transplants there must therefore be ABO blood group compatibility between the recipient and the donor organ, using the same rules for blood transfusion (*Table 88.3*).

TABLE 88.3 Blood group compatibility.

Donor blood group	Permissible recipients
O (universal donor)	O, A, B and AB
A	A and AB
B	B and AB
AB (universal recipient)	AB

HLA matching

Allograft rejection is directed against human leukocyte antigens (HLAs). These are a group of cell surface glycoprotein molecules. HLA molecules are divided into class I (A, B and C) and class II (DR, DP and DQ). Class I molecules are expressed on the surface of all nucleated cells, but class II are only expressed by antigen-presenting cells (APCs) such as dendritic cells and B lymphocytes. HLA molecules are encoded for in the major histocompatibility complex (MHC) on chromosome 6. They are highly polymorphic, i.e. their amino acid sequences differ widely between individuals. To give an example, there are >1000 variants of the HLA-B gene. This genetic variability means that most transplant donors and recipients have different HLA profiles. Donors and transplant recipients are HLA-typed using DNA sequencing. The antigens at HLA-A, -B and -DR are together described as the tissue type.

The level of mismatch between the HLA molecules determines the strength of the immune response. Every individual has two copies of each HLA gene and so for each locus (A, B or DR) it is possible to have 0, 1 or 2 mismatched genes. As the number of mismatches increases so does the chance of immune recognition and rejection. The best matched grafts are described as having a mismatch of 0–0–0, i.e. no mismatches at A, B or DR, respectively. A completely mismatched graft would be annotated as a 2–2–2 mismatch. The effects of the different HLA antigens are not uniform. DR mismatches have a more powerful effect than B mismatches, and A mismatches are the least important. The tissue types of potential renal transplant recipients are held by national organisations, such as NHS Blood and Transplant in the UK, and used to allocate donor organs to patients with the lowest level of HLA mismatch.

The immune response to a transplanted organ

The main immune cells involved in transplant immunology are APCs and T and B lymphocytes. These cell types interact by a series of specific surface molecules that are designated by CD (cluster of differentiation) numbers. Cell surface receptors can only interact if they have the correct complementary three-dimensional structures.

The immune response to an allograft is orchestrated by T lymphocytes. When T cells encounter foreign transplant HLA antigens they become activated and then proliferate into clones of cells that attack the allograft. Full T-cell activation requires a number of signals (*Figure 88.6*). Transplant antigen

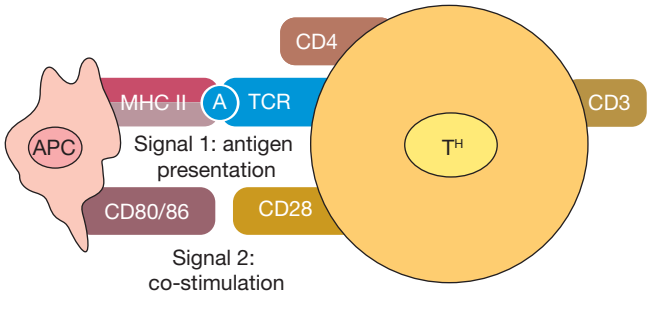

Figure 88.6 Interaction between an antigen-presenting cell and a CD4+ T-helper cell. A, antigen; APC, antigen-presenting cell; CD, cluster of differentiation; MHC II, major histocompatibility complex class II; TCR, T-cell receptor; T^H, T-helper cell. (Adapted with permission from Clatworthy M, Watson C, Allison M, Dark J. *Transplantation at a glance*. John Wiley and Sons Ltd, 2012.)

is presented to T-helper cells (CD4+) on self-MHC class II molecules by specialised APCs. The class II/peptide antigen complex is recognised by the T-cell receptor (signal 1). CD4+ T-helper cell activation also requires a co-stimulatory signal involving the interaction of pairs of molecules, one on the surface of the T cell and the other on the surface of the APC (signal 2). An example of signal 2 is the interaction of T-cell CD28 with CD80 on the APC. When both signal 1 and signal 2 are received the CD4+ T-helper cell will upregulate expression of the interleukin-2 (IL-2) receptor (CD25) on its cell surface. Binding of IL-2 to its receptor causes further activation of the T cell (signal 3). The T cell will then proliferate and release more IL-2, which, in turn, leads to activation and clonal proliferation of T-killer cells (CD8+). These cytotoxic killer cells infiltrate the allograft and cause cell death by the release of molecules called perforin and granzyme. Perforin punches holes in the target cell membrane and this allows passive diffusion of granzyme into the cell, where it activates caspase enzymes and causes cell death by apoptosis.

Antigen presentation in transplantation

There are two main types of antigen presentation to T lymphocytes (*Figure 88.7*). Direct antigen presentation involves donor APCs showing intact and unprocessed donor HLA (class I or class II) molecules on their cell surface to recipient T cells. In contrast, indirect presentation is performed by recipient APCs. These internalise foreign donor HLA molecules from the graft, process them into short peptide fragments and then load them into the peptide groove of recipient (self) HLA class II molecules. The donor peptide–recipient HLA class II complex is then expressed on the cell surface and presented to recipient T cells. Direct antigen presentation only occurs for the first 6–12 weeks after transplantation because this is the lifespan of the donor passenger APCs that are present in the allograft. As indirect presentation involves recipient APCs it is a long-lived response.

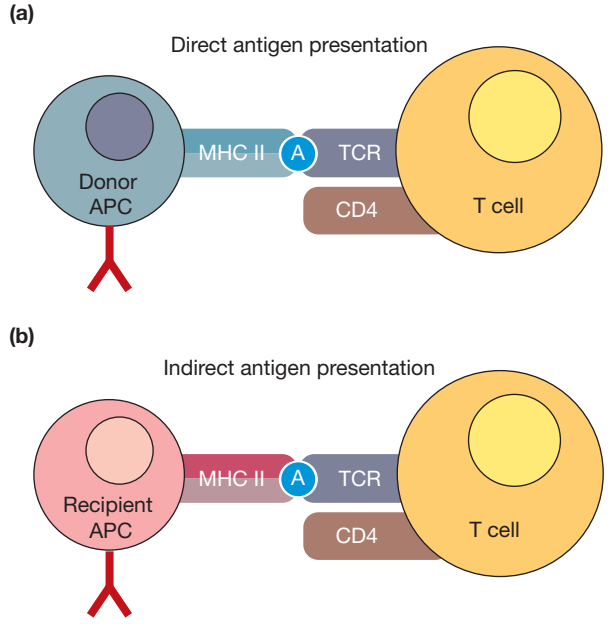

Figure 88.7 Direct (a) and indirect (b) antigen presentation. A, antigen; APC, antigen-presenting cell; CD, cluster of differentiation; MHC II, major histocompatibility complex class II; TCR, T-cell receptor. (Adapted with permission from Clatworthy M, Watson C, Allison M, Dark J. *Transplantation at a glance*. John Wiley and Sons Ltd, 2012.)

TRANSPLANT REJECTION

Allograft rejection can be divided into distinct types.

Hyperacute rejection

Hyperacute rejection is extremely rare. It can result from an inadvertent ABO blood group-incompatible transplantation after a clerical error in recording blood groups. It may also occur when there are preformed circulating donor-specific HLA antibodies (DSA) or ABO antibodies. In either case, immediately after revascularisation of the transplant antibodies will bind to ABO or HLA antigens on the surface of the graft vascular endothelial cells. The antibodies are complement fixing and this leads to activation of the complement cascade and phagocytes. This leads to endothelial damage with release of tissue factor and activation of the clotting cascade. Widespread intravascular thrombosis causes infarction of the allograft within minutes or hours. This is untreatable and inevitably results in graft loss.

Acute cell-mediated rejection

In the era of modern immunosuppressive drugs, the incidence of acute cell-mediated rejection (CMR) is only 10–20%. Acute CMR is largely mediated by direct antigen presentation. As donor APCs in the allograft have a lifespan of only a few weeks, the peak incidence of acute CMR is in the first 3 months post transplantation. The characteristic biopsy finding of acute CMR is a marked interstitial lymphocytic infiltrate (*Figure 88.8*). In the kidney, the presence of lymphocytes inside the basement membrane of the renal tubular epithelium is referred to as tubulitis and is diagnostic of acute CMR. Initial treatment is by high-dose pulsed intravenous steroids (methylprednisolone 0.5 g intravenously for 3 days), which is

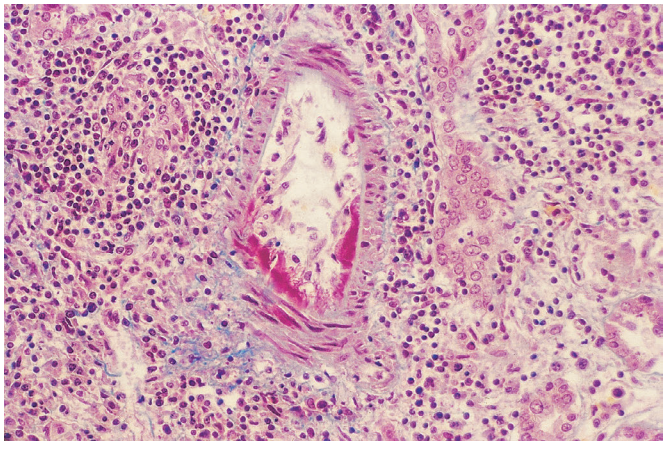

Figure 88.8 Severe acute renal allograft rejection with a heavy mononuclear cell infiltrate and intimal arteritis.

successful in up to 90% of cases. Severe or steroid-resistant rejection is treated with lymphocyte-depleting intravenous antithymocyte globulin (ATG).

Acute antibody-mediated rejection

Acute antibody-mediated rejection (AMR) occurs in <5% of renal transplants but is more serious and more difficult to treat than cell-mediated rejection. It is caused by HLA DSA, which are produced by a sensitising episode from a previous transplant, a blood transfusion or pregnancy. Binding of DSA to mismatched HLA antigens on the surface of allograft endothelial cells leads to activation of the complement system and tissue injury. In the kidney, transplant biopsy will show inflammation of the vessels (vasculitis) and deposition of the complement component C4d in the peritubular capillaries (*Figure 88.9*). Treatment is by plasma exchange to remove circulating antibodies. Rituximab, an anti-CD20 monoclonal antibody, can also be used to destroy B cells and prevent further production of DSA.

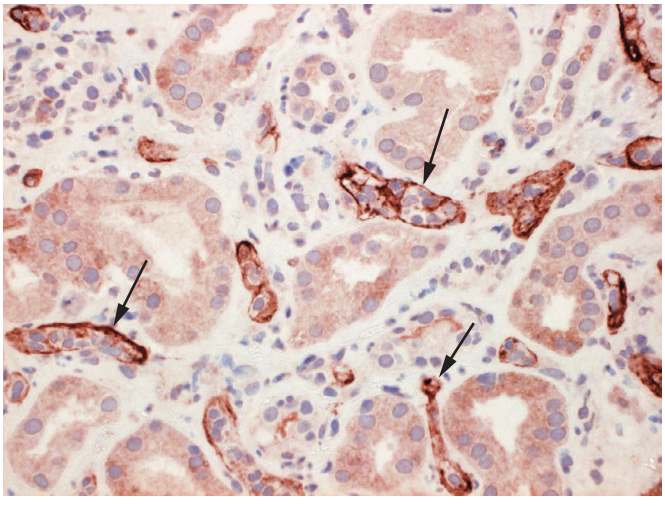

Figure 88.9 Acute antibody-mediated renal allograft rejection. There is widespread staining for the complement component C4d within the peritubular capillaries (arrows), which indicates alloantibody binding to the graft vasculature.

Chronic antibody-mediated rejection

The pathophysiology is not well understood but the long-lived indirect antigen presentation pathway is likely to be important. CD4 T-cell activation also promotes donor-specific alloantibody production, and this causes ongoing allograft damage.

IMMUNOSUPPRESSION

Modern immunosuppression is so effective that acute rejection rates of 10–20% can be achieved in all types of solid organ transplantation. The challenge is to deliver sufficient immunosuppression to prevent rejection while minimising drug side effects. Immunosuppression also increases the risk of both infection and malignancy. This is a non-specific effect related to the total burden of immunosuppression rather than agent-specific side effects.

The mechanisms of action of current immunosuppressive drugs depend on anti-inflammatory effects and the prevention of lymphocyte activation and proliferation (*Figure 88.10*). Lymphocytes are some of the most rapidly dividing cells in the body and their activation and clonal expansion forms the immunological basis of acute allograft rejection.

Immunosuppression for transplantation has two phases: induction and maintenance. Induction therapy commonly consists of a combination of high-dose intravenous steroids and the anti-CD25 monoclonal antibody basiliximab, which blocks IL-2 receptors. Other induction agents for high-risk cases are ATG and the monoclonal antibody alemtuzumab (Campath).

The commonest maintenance immunosuppressive regimen for solid organ transplants consists of triple therapy with a calcineurin inhibitor (ciclosporin or tacrolimus), an antiproliferative agent (azathioprine or mycophenolic acid) and steroids.

Calcineurin inhibitors (ciclosporin and tacrolimus)

The calcineurin inhibitors (CNIs) are the mainstay of modern immunosuppressive regimens. CNIs prevent transcription of the IL-2 gene in T cells. As IL-2 is the main T-cell growth factor, inhibition of its production prevents T-cell proliferation.

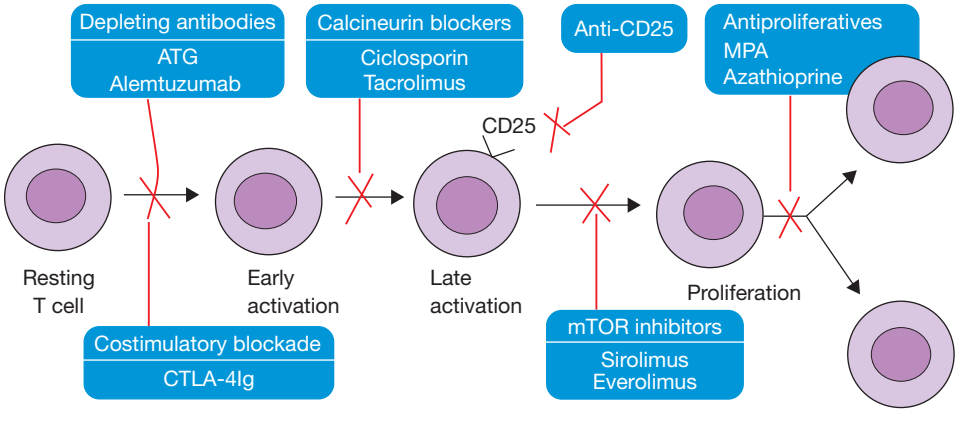

Figure 88.10 Site of action of immunosuppressive agents on T cell. ATG, antithymocyte globulin; CTLA-4Ig, cytotoxic T-lymphocyte-associated protein 4 immunoglobulin; MPA, mycophenolic acid derivatives; mTOR, mammalian target of rapamycin.

The main side effect of both ciclosporin and tacrolimus is nephrotoxicity. CNIs have a narrow therapeutic index (a small difference between the minimum effective concentration and the minimum toxic concentration). CNI dosage must therefore be guided by monitoring drug blood levels.

Antiproliferative agents (azathioprine and mycophenolic acid)

These drugs are antiproliferative agents. Their mechanism of action is to block purine nucleotide synthesis. This prevents replication of DNA and thus interferes with lymphocyte proliferation. The main side effects of both drugs are bone marrow suppression (anaemia, leukopenia, thrombocytopenia) and gastrointestinal symptoms (nausea, vomiting and diarrhoea).

Corticosteroids

These are potent anti-inflammatory agents that have wide-ranging effects on the immune system. Prolonged exposure to steroids causes numerous potential side effects, including: a Cushingoid appearance with a moon face, central obesity, abdominal striae and proximal myopathy; thin skin that bruises easily; hypertension; glucose intolerance that may lead to new-onset diabetes mellitus; osteoporosis; and peptic ulcer disease. These may be minimised by rapid reduction or withdrawal of steroids.

Mammalian target of rapamycin (mTOR) inhibitors

The mTOR inhibitors rapamycin and everolimus act by binding to and inhibiting a cytoplasmic kinase enzyme complex called mTOR. This prevents intracellular signalling from the IL-2 receptor. The downstream effect is arrest of T-cell division at the G1–S phase. The cell cycle effects of mTOR inhibitors are not limited to lymphocytes and their side-effect profile includes severe mouth ulceration, poor wound healing and lymphocele formation. This limits the use of mTOR inhibitors in the first few weeks post transplantation. However, mTOR inhibitors are not nephrotoxic and they may be used 3 months or more after transplantation as an alternative to CNIs to minimise CNI-associated renal dysfunction.

KIDNEY TRANSPLANTATION

End-stage renal disease

The incidence of end-stage renal disease (ESRD) in the UK is approximately 120 per million population and around 8000 people require renal replacement therapy annually. The leading causes of ESRD are diabetes, hypertension and chronic glomerulonephritis (*Summary box 88.1*). ESRD is largely a disease of older adults with the mean age at commencement of renal replacement therapy being 64 years.

Summary box 88.1

Common causes of ESRD

- Chronic glomerulonephritis
- Diabetic nephropathy
- Hypertension
- Renal vascular disease
- Polycystic kidney disease
- Chronic pyelonephritis
- Obstructive uropathy

Rationale for kidney transplantation

Kidney transplantation improves life expectancy and quality of life when compared with dialysis. However, only approximately one-third of patients with ESRD are fit enough to withstand transplant surgery and long-term immunosuppression. Successful transplantation frees patients from the rigors of dialysis and eliminates uraemic symptoms. Transplant kidneys produce normal levels of erythropoietin and this reverses the anaemia of chronic renal disease. Transplant patients therefore have more energy and better exercise capacity than patients on dialysis. There are also no fluid or dietary restrictions after transplantation. Importantly, for women of child-bearing age pregnancy is also possible after a successful kidney transplant. Life expectancy is higher in the transplant population with 5-year survival of >85% compared with <50% for patients on dialysis. However, these figures cannot be directly compared as there is selection bias because only relatively fit patients are offered transplantation.

Selection of patients for transplantation

Potential transplant recipients undergo a rigorous work-up process to identify major comorbidities that would preclude transplant surgery. Age per se is not a contraindication to renal transplantation and it is now common to transplant patients in their seventies as long as they have the necessary cardiovascular fitness. Uncontrolled infection and most malignancies are contraindications to transplantation. Patients with ESRD have a greatly increased risk of cardiovascular disease and require a chest radiograph, electrocardiogram (ECG) and, if indicated, an echocardiogram. Patients with a history of diabetes or ischaemic heart disease should undergo a stress echocardiogram and sometimes coronary angiography.

There are a number of technical considerations before a patient is deemed suitable for transplantation. The iliac blood vessels must be suitable for anastomosis and there must be a means of draining the transplant ureter. In patients with a history of vascular disease or deep venous thrombosis, the iliac arterial and venous system should be assessed by Doppler ultrasound scanning and possibly also MRA or CT angiography. Patients who have been anuric for a number of years may have a small non-compliant bladder and this should be assessed by urodynamics. In patients with polycystic kidney disease, the size of the native kidneys should be assessed clinically to make sure that there is sufficient room for a kidney

Harvey Williams Cushing, 1869–1939, Professor of Surgery, Harvard University Medical School, Boston, MA, USA.
Christian Johann Doppler, 1803–1853, Professor of Experimental Physics, Vienna, Austria, enunciated the Doppler principle in 1842.

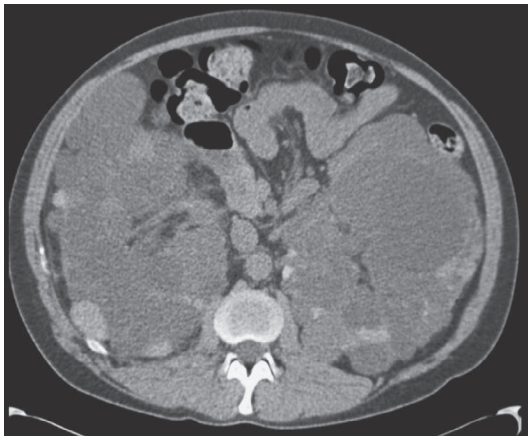

Figure 88.11 Computed tomography scan of a patient with very large polycystic kidneys that extend well into both iliac fossae. The patient requires removal of one of the polycystic kidneys to make room for a subsequent renal transplant. Nephrectomy should be performed several weeks before transplantation.

transplant in at least one iliac fossa. If there is inadequate space on both sides, then a pretransplant native nephrectomy will be necessary (*Figure 88.11*). With the exclusion of non-melanotic skin cancer, patients who have had a malignant disease should be deferred for a disease-free period of at least 2 years. Where possible patients should be listed for transplantation pre-emptively when they are within 6 months of requiring dialysis. This equates to an estimated glomerular filtration rate (eGFR) of 10–15 mL/min/1.73m^2.

RENAL TRANSPLANT SURGERY

Preparation of the donor kidney

The donor kidney must be examined and prepared on the back-table in order to check that it is suitable to be transplanted. Multiple renal arteries are present in up to 25% of kidneys and, if present, may require reconstructive bench surgery to simplify implantation. The renal arteries are end arteries,

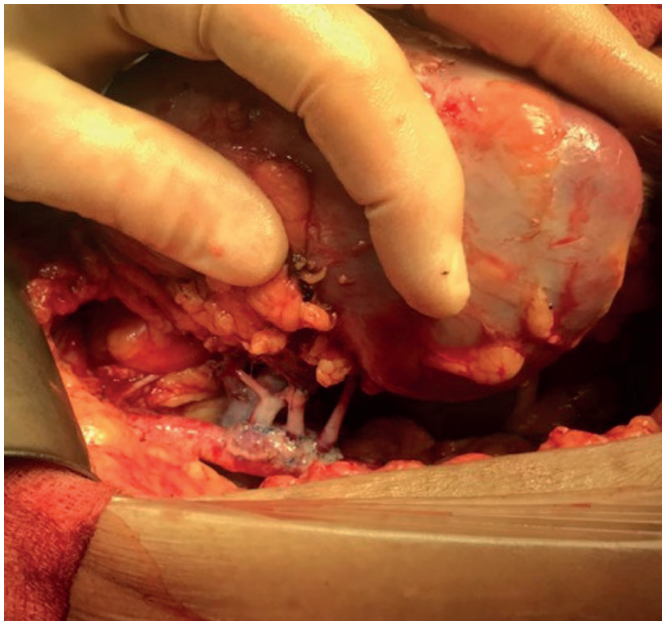

Figure 88.12 Deceased donor renal transplant with three arteries that have been reconstructed using a patch of donor aorta and anastomosed to the external iliac artery.

so it is important whenever possible to preserve all branches (*Figure 88.12*).

Renal transplant operative technique

The donor kidney is transplanted heterotopically into one of the iliac fossae via a curvilinear incision. The peritoneum should be kept intact and swept upwards to reveal the iliac vasculature. The transplant renal vein is anastomosed end to side to the external or common iliac vein. The renal artery is anastomosed either end to side to the external or common iliac artery or end to end to the divided internal iliac artery (*Figure 88.13*). The internal iliac artery is used more commonly for live donor kidneys because of the lack of an aortic patch. There are several

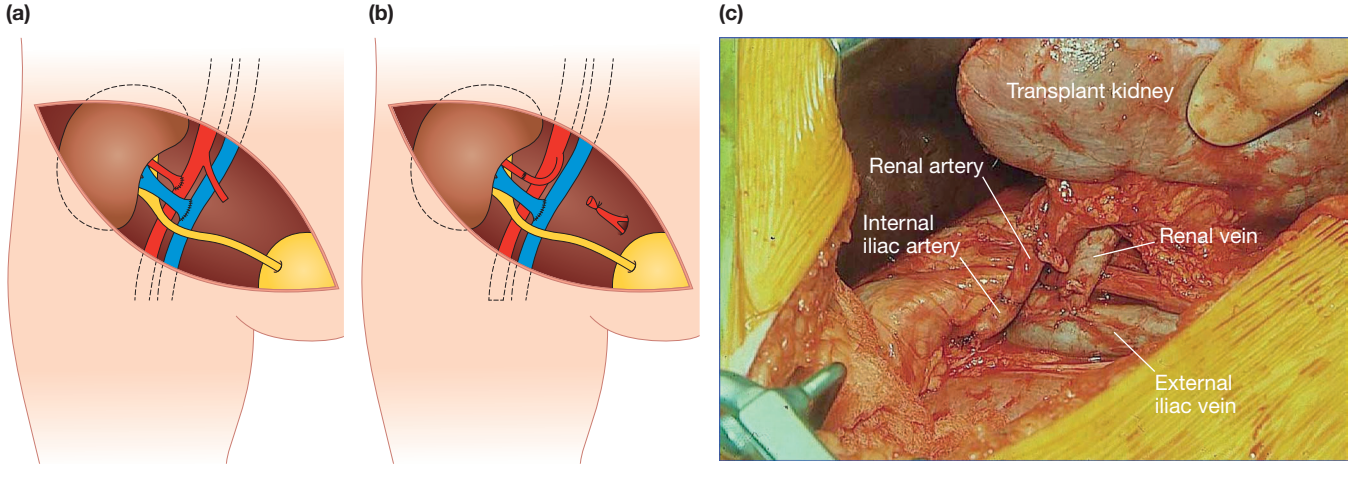

Figure 88.13 Renal allograft implantation techniques. (a) Anastomosis of the renal artery on a donor aortic patch end to side to the external iliac artery. (b) Anastomosis of the renal artery end to end to the divided internal iliac artery. (c) Operative photograph of anastomosis of the renal artery end to end to the internal iliac artery and the renal vein end to side to the external iliac vein.

techniques for anastomosing live donor kidneys with multiple arteries, but it is best to minimise the number of anastomoses by careful bench surgery. For example, equal-sized arteries can be 'trousered' to create a single ostium for anastomosis. Upper or lower polar arteries may also be anastomosed to the divided inferior epigastric artery. It is especially important to anastomose lower polar arteries as these may provide the only blood supply to the ureter. After revascularisation of the transplant kidney the ureter is anastomosed to the bladder as an extravesical onlay (the Lich–Grégoir technique) (*Figure 88.14*). It is now routine practice to place a double-J stent across the ureteric anastomosis. The stent is removed by flexible cystoscopy under local anaesthesia after a few weeks. Before closing the wound it is important to ensure that the kidney is lying in a satisfactory position without kinking of the renal blood vessels.

Dual kidney transplantation

This involves the transplantation of a pair of marginal quality kidneys from the same donor into one recipient in order to provide adequate nephron mass. Both kidneys can be placed in the same iliac fossa. This approach is used for kidneys from elderly DCD donors and so-called expanded criteria donors, which are defined by age >60 years or age >50 years with at least two of the following: hypertension; terminal creatinine >133 μmol/L; death from stroke.

Renal transplantation in children

In children with established renal failure, kidney transplantation facilitates their growth and development and markedly improves the quality of life of the child and their parents. In young children there is often a size mismatch between the donor kidney, which may be from an adult, and the recipient. In this case the kidney is transplanted intraperitoneally with anastomosis of the renal artery to the distal aorta and the renal vein to the inferior vena cava or common iliac vein.

Living donor kidney transplantation

This accounts for approximately 1000 kidney transplants annually in the UK, which is approximately one-third of the total renal transplant programme. Living donors may be related, unrelated, altruistic or part of a donor exchange scheme (for ABO blood group or HLA incompatibility). Potential live donors undergo extensive assessment that includes urine and blood screening tests, chest radiograph, ECG, echocardiography (if indicated) and CT angiography or MRA for the assessment of renal vascular anatomy. Renal function is measured by an accurate isotope GFR technique and must be above evidence-based age- and gender-specific safety thresholds for donation. Following unilateral nephrectomy, there is considerable compensation by the remaining kidney and donors are usually left with 70% of their predonation GFR. The mortality from donor nephrectomy is approximately 1:3000 and major morbidity occurs in <5%. Donation increases the risks of hypertension, renal failure and pregnancy-related complications.

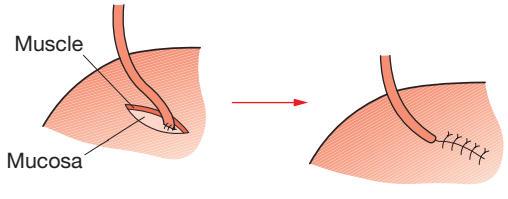

Figure 88.14 Ureteric implantation by direct anastomosis to a small cystotomy: Lich–Grégoir technique.

Minimally invasive donor nephrectomy

Laparoscopic surgery is now widely established and there are a number of techniques. After a fully laparoscopic transperitoneal dissection (*Figure 88.15*), the kidney is removed through a small retrieval incision. Hand-assisted laparoscopic nephrectomy is also used widely. A hand port is used to aid the dissection and for extraction of the kidney. This technique is easier to learn and can be safely performed by surgeons with less laparoscopic experience.

Immunosuppressive regimen for renal transplantation

Immunosuppression for renal transplantation generally comprises induction therapy with the anti-CD25 monoclonal antibody basiliximab followed by a maintenance regimen of a calcineurin inhibitor (most usually tacrolimus), an antiproliferative agent (usually mycophenolic acid) and corticosteroid (prednisolone). The anti-CD52 monoclonal antibody alemtuzumab can be used for induction instead of basiliximab for

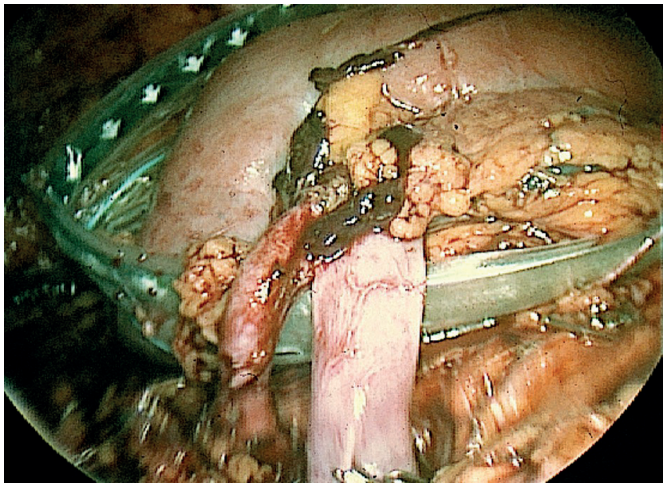

Figure 88.15 Completed laparoscopic dissection of the left kidney, which is ready for stapling of the vessels and removal in an Endocatch bag.

Robert Lich Jr, urologist, Louisville, KY, USA.
Willy Grégoir, Chef de Clinique Urologique, Brussels, Belgium (1962–1987).

patients at high risk of allograft rejection or as part of a steroid avoidance strategy in patients at high risk of developing diabetes post transplantation.

Early postoperative course

Accurate fluid and electrolyte balance are maintained with the help of central venous pressure monitoring. Hyperkalaemia is common in the early post-transplant period, especially in patients with DGF. This should be managed initially with intravenous glucose and insulin but early dialysis is often required. Recovery is straightforward in the majority of renal transplant patients. The bladder catheter is removed on postoperative day 5 and most patients can be discharged on day 6.

Delayed renal allograft function

DGF is defined as the need for dialysis in the first 7 days post transplant. The patient will be oliguric or anuric and the serum creatinine will fail to fall in the early postoperative period. DGF is usually the clinical consequence of acute tubular necrosis related to ischaemia–reperfusion injury. It occurs in around 30% of DBD kidneys and 60% of DCD kidneys because of the significantly longer warm ischaemic period prior to *in situ* organ cooling. In contrast, DGF occurs in <5% of live donor kidneys because of the short cold ischaemic time. An early Doppler ultrasound study should be performed in all patients with DGF to check that the graft is well vascularised. The management of DGF is supportive with haemodialysis, careful fluid balance and avoidance of CNI toxicity. Graft function usually recovers within a few days, but it may take several weeks. Primary non-function is the term used for grafts that never work. This occurs in <5% of renal transplants as a result of either vascular thrombosis or irreversible ischaemic injury leading to cortical necrosis.

Surgical complications of renal transplantation

Haemorrhage

A haematoma may develop in the transplant bed in the first few postoperative days. This is often due to bleeding from small unsecured vessels in the renal hilum that were not apparent at the time of surgery. In this situation the patient is haemodynamically stable and can be investigated by CT angiography and subsequently re-explored to remove the haematoma.

Anastomotic haemorrhage can also occur but is very rare. It presents as haemorrhagic shock associated with a fall in the haemoglobin to <50 g/L. This is a life-threatening situation and the patient should be returned to the operating theatre immediately. It may be possible to repair a defect in an arterial or venous anastomosis but transplant nephrectomy is often required.

Late anastomotic haemorrhage can also occur after some weeks or months owing to the development of a mycotic aneurysm. The commonest causative organism is donor-derived *Candida albicans*. A ruptured mycotic aneurysm requires an immediate graft nephrectomy and has a significant mortality.

Renal artery thrombosis

Although this occurs in <1% of renal transplants, the most common outcome is loss of the allograft. Most cases are due to technical complications at the arterial anastomosis. The usual presentation of renal artery thrombosis is sudden anuria and a rapid decline in renal function. The diagnosis may be missed in patients who have DGF or a normal urine output from their native kidneys. An urgent Doppler ultrasound scan will show an absence of perfusion to the graft. The patient must be returned to the operating theatre immediately to attempt thrombectomy, but most grafts will already be infarcted and transplant nephrectomy will be required.

Renal vein thrombosis

Renal vein thrombosis occurs in 1–5% of transplants. It may be due to a technical error such as kinking or torsion of the venous anastomosis, but many cases are idiopathic. The peak incidence occurs at postoperative days 3–7. The presentation is distinctive with sudden pain over the renal transplant associated with frank haematuria. In early cases with partial thrombosis, Doppler ultrasound scanning may demonstrate reversal of arterial blood flow in diastole. In an established renal vein thrombosis Doppler scanning will demonstrate marked swelling of the renal allograft, often with a significant surrounding haematoma due to rupture of the renal capsule. Re-exploration to attempt a thrombectomy is rarely successful and the vast majority of cases lead to graft loss.

Urological complications

Urological complications occur in approximately 5% of renal transplants but they rarely result in graft loss. After dissection of the ureters at organ retrieval, only the ureteric blood supply from the renal artery is preserved. In consequence, the blood supply of the distal transplant ureter can be poor.

Urinary leak

Urinary leaks can occur in any part of the urinary drainage system but are commonly due to ischaemic necrosis of the distal ureter. The peak incidence is at the time of urinary catheter removal (day 5) but leaks can be delayed for a few weeks. In early leaks clear fluid discharges through the wound or collects in the drains. Biochemical analysis will show that the fluid is urine (creatinine in the millimolar range) rather than lymph (creatinine in the micromolar range and equal to the serum level). Leaks presenting later on when the wound is fully healed present as a peritransplant fluid collection.

Initial management is bladder catheterisation and cystography to confirm the diagnosis. The anatomical site and extent of the leak can be determined by inserting a percutaneous nephrostomy and performing antegrade pyelography. It is sometimes possible to place a double-J stent across the leakage point using an antegrade approach. If there has been significant ischaemic necrosis of the distal transplant ureter the leak will not resolve and surgery will be required. There are a number of alternatives for reconstructing the urinary drainage system. If there is a sufficient length of transplant ureter after excision of the necrotic segment, the ureter can be reimplanted directly into the bladder over a new double-J stent. If the

healthy section of the transplant ureter is too short, a native-to-transplant ureteroureterostomy or ureteropyelostomy can be performed. Finally, if the ipsilateral native ureter is absent, because of a previous nephrectomy, a Boari bladder flap can be fashioned to drain the transplant kidney.

Ureteric obstruction

This can occur at any time after transplantation (days to years). Early obstruction is usually due to a technical error in the bladder anastomosis. Obstruction occurring after 3 months is invariably due to an ischaemic stricture. BK polyomavirus infection is also a cause of ureteric stricture. Patients present with renal dysfunction and investigation by ultrasound scanning reveals hydronephrosis. The initial management of choice is to place a percutaneous nephrostomy. An antegrade pyelogram can then be performed to define the site and extent of the stricture. Short strictures may be treated by interventional radiology with antegrade balloon dilatation and placement of a double-J stent across the stricture. Long, tight strictures require surgery and the options for reconstruction are the same as those for a urine leak.

Lymphocele

The incidence of significant lymphoceles (>3 cm) that cause complications and need treatment is around 5%. Small lymphoceles (<3 cm) are more common but asymptomatic and resolve spontaneously. The usual source of a lymph leak is from lymphatics that are divided during dissection of the iliac vessels. Lymph accumulates as a collection because of the extraperitoneal position of the kidney. Large lymphoceles can compress surrounding structures and, in some cases, cause renal vein thrombosis or ureteric obstruction.

All large peritransplant collections should be aspirated under ultrasound to exclude a urinary leak. Lymph is characterised by having the same biochemical profile as serum. Percutaneous drainage, sometimes on a number of occasions, leads to resolution of most lymphoceles. If this fails surgical drainage is required. This is usually performed laparoscopically and involves creating a fenestration in the wall of the lymphocele so that it drains into the peritoneal cavity.

Transplant renal artery stenosis

The incidence of transplant renal artery stenosis (TRAS) is approximately 5%. The usual presentation is refractory hypertension associated with allograft dysfunction at 3–6 months post transplant. TRAS activates the renin–angiotensin–aldosterone system and may become evident by a sudden increase in serum creatinine after prescribing an angiotensin-converting enzyme inhibitor.

Doppler scanning is suggestive of TRAS when the arterial peak systolic velocity is >250 cm/s and the waveform may have the characteristic tardus parvus (Latin for late and small) pattern. MRA or CT angiography are useful for anatomical definition. Percutaneous transluminal angioplasty with or without stenting is the preferred treatment.

Acute renal allograft dysfunction

This can be defined as a rise in serum creatinine of >10% from baseline or an absolute rise of ≥20 μmol/L. Common causes are listed in *Summary box 88.2*. Allograft dysfunction should be investigated by urinalysis and culture to exclude urinary tract infection, CNI blood levels for drug-mediated nephrotoxicity and Doppler ultrasound scanning to check for arterial perfusion, venous drainage and hydronephrosis due to urinary obstruction. If there is uncertainty about the cause of graft dysfunction, a percutaneous needle-core transplant biopsy should be performed to establish whether allograft rejection is present.

Summary box 88.2

Causes of early graft dysfunction

Any rise in serum creatinine of >10% of baseline or ≥20 μmol/L should be considered as acute allograft dysfunction that requires investigation. Possible causes are:

- Acute rejection (antibody mediated or cell mediated)
- Calcineurin inhibitor toxicity
- Dehydration
- Urinary tract infection or pyelonephritis
- Any other source of sepsis
- Renal vein or renal artery thrombosis
- Ureteric obstruction or urine leak

Long-term allograft dysfunction

There are immunological and non-immunological causes for a progressive decline in renal allograft function over a period of months or years (*Summary box 88.3*). The commonest cause of long-term renal allograft damage and subsequent loss is chronic AMR. This is often related to inadequate compliance with immunosuppressive medication. There is no proven treatment for chronic AMR.

Non-immunological causes are also significant risk factors for chronic allograft damage. Irrespective of the cause, chronic transplant injury leads to interstitial fibrosis and tubular atrophy. Once established these changes are irreversible and will eventually lead to failure of the transplant.

The two most important factors in maintaining good long-term allograft function are meticulous adherence to immunosuppressive medication and good blood pressure control (<130/80 mmHg).

Outcomes after renal transplantation

Patient and graft survivals vary according to donor type and are presented in *Table 88.4*. Overall, kidney transplantation is a highly successful treatment for ESRD. Living donor transplants yield better results than deceased donor transplants.

Achille Boari, nineteenth century Italian urological surgeon from Ferrara, described the technique of a bladder flap in dogs in 1894; it was first performed in a patient in 1936.

Summary box 88.3

Causes of long-term graft dysfunction

- Immunological
  - Chronic AMR
  - Acute CMR or AMR, which can occur at any time
- Non-immunological
  - Pre-existing damage in the donor kidney (especially relevant to DCD and extended criteria donor kidneys)
  - Early ischaemia–reperfusion injury
  - Chronic calcineurin nephrotoxicity
  - Ureteric or bladder outflow obstruction
  - Recurrent urinary tract infection or pyelonephritis
  - BK polyomavirus nephropathy
  - Recurrent native disease: glomerulonephritis; focal segmental glomerulosclerosis; immunoglobulin A nephropathy
  - Renal artery stenosis
  - Poorly controlled hypertension
  - Dyslipidaemia

DBD and DCD kidney transplants have comparable patient and graft survival.

Although these survival analyses are helpful, most patients want to know how long their transplant can be expected to last. This is best expressed by graft half-life ($t_{½}$) data (median graft survival). The current overall $t_{½}$ for live donor transplants is approximately 25 years and for deceased donor transplants approximately 15 years.

For all types of kidney, increasing donor age is the most important risk factor for poorer graft survival. Long cold ischaemic time (>12 hours for DCD kidneys and >24 hours for DBD kidneys) is also associated with reduced graft survival. Graft survival also decreases as the number of HLA mismatches increases.

TABLE 88.4 Patient and graft survival after renal transplantation.

Type of allograft (donor source)	5-year patient survival	10-year patient survival	5-year graft survival	10-year graft survival
Live donor	95%	90%	90%	80%
Deceased donor	90%	80%	85%	75%

Deceased donor includes both DBD and DCD results, which are comparable.

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